

Deep Learning-Based IDH Mutation Status Prediction in Gliomas: Evaluating the Contribution of ASL Perfusion MRI

by

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ABSTRACT

Gliomas are the most common malignant primary brain tumors in adults. The 2021 World Health Organization (WHO) classification makes isocitrate dehydrogenase (IDH) mutation status a central part of glioma diagnosis, as IDH-mutant and IDH-wildtype tumors behave differently. Knowing the IDH status before treatment is important, but it is currently determined through invasive tissue sampling. Deep learning applied to MRI offers a non-invasive alternative, but most existing approaches rely on contrast-enhanced imaging (using gadolinium), which has its own clinical drawbacks. This project investigates whether IDH mutation status can be predicted entirely from contrast agent-free brain MRI. We use three contrast agent-free sequences (T2w, FLAIR, and ASL) and a 2D ResNet-18 model that is pre-trained on ImageNet. We compare a plain (no-mask) pipeline with a soft-mask pipeline that highlights the tumor while keeping the whole-brain context, and we measure how much the ASL (perfusion) sequence contributes. On the UCSF-PDGM dataset, the best model (soft-mask, $\alpha = 0.5$) reaches a test AUC of 0.987. Removing ASL lowers the AUC, which shows that perfusion carries useful information. As these results are based on a single data split, they should be considered preliminary.

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1 Introduction

1.1 Background and Motivation

Gliomas are the most common malignant primary brain tumors in adults [1]. In 2021, the World Health Organization (WHO) updated its classification of central nervous system tumors and made molecular markers central to the diagnosis [1]. The most important of these markers is the isocitrate dehydrogenase (IDH) mutation. IDH-mutant and IDH-wildtype gliomas grow differently, respond to treatment differently, and have very different prognoses, so knowing the IDH status is important for planning a patient’s care.

Currently, definitive IDH status is determined through invasive tissue sampling (biopsy or surgery). However, this is invasive and not always possible before treatment decisions are made. As a non-invasive alternative, researchers use deep learning applied to magnetic resonance imaging (MRI) to predict IDH status, but most existing methods rely on gadolinium-based contrast agents. Contrast agents add cost, cannot be given to some patients (for example, patients with kidney problems), and there are concerns about gadolinium staying in the body. Because of these combined drawbacks, a method that can predict IDH status non-invasively and without contrast agents would be highly valuable.

1.2 Objectives and Contributions

The main goal of this project is to build and test a model that predicts IDH mutation status from contrast agent-free brain MRI. The project has three specific objectives:

1. Build a classifier that predicts IDH mutation status using only contrast agent-free MRI sequences (T2w, FLAIR, and ASL) with a 2D ResNet-18 model.
2. Test whether focusing the model on the tumor with a soft mask improves performance, by comparing a no-mask pipeline with a soft-mask pipeline.
3. Measure how much the ASL (perfusion) sequence contributes to the prediction.

Overall, this work shows that contrast agent-free MRI can be used to predict IDH mutation status non-invasively, and it provides evidence about the role of tumor focusing and ASL perfusion in this task.

1.3 Report Organization

The rest of this report is organized as follows. Section 2 gives background on the MRI sequences and reviews related work. Section 3 describes the dataset and the methods. Section 4 presents the results. Section 5 concludes the report with a discussion, future work, and realistic constraints.

2 Background and Related Work

2.1 MRI Sequences and ASL Perfusion

This project uses three contrast agent-free MRI sequences. T2-weighted (T2w) and Fluid-Attenuated Inversion Recovery (FLAIR) images are anatomical sequences that show the tumor and the surrounding tissue, such as edema. Arterial Spin Labeling (ASL) is a perfusion sequence: it uses the water in the blood as a natural tracer to measure blood flow in the brain without any injected contrast agent.

ASL is interesting for this task because tumor blood flow is linked to tumor biology. Studies have reported that IDH-mutant gliomas tend to have lower perfusion than IDH-wildtype gliomas [6, 9]. Since ASL measures perfusion directly and without contrast, it may carry useful information about the IDH status.

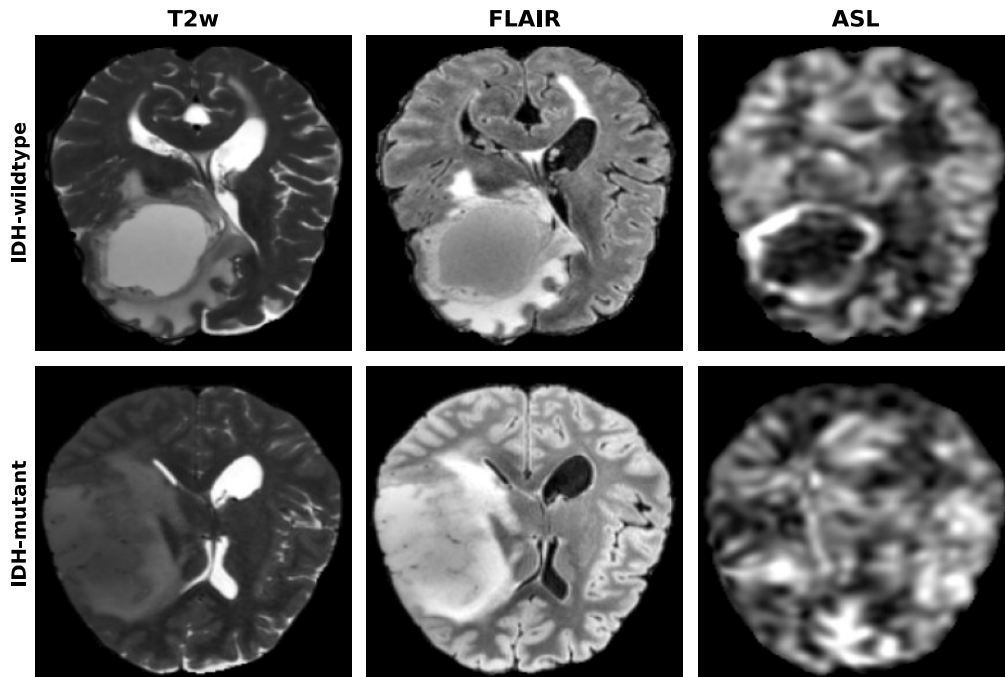


Figure 2.1: Example contrast agent-free MRI sequences (T2w, FLAIR, ASL) for an IDH-wildtype (top) and an IDH-mutant (bottom) glioma.

2.2 Related Work

Predicting molecular markers of gliomas from MRI has been an active research area. Many studies use radiomics (hand-crafted image features with classical machine learning) or deep learning to predict IDH status from MRI [6, 7]. Convolutional neural networks, and ResNet models in particular, have been used for this task. Most of these works rely on anatomical sequences, and some also use perfusion imaging.

This project builds on the work of our research group. Buz-Yalug et al. used dynamic susceptibility contrast (DSC) perfusion MRI with deep learning to identify IDH and TERTp mutations [6]. Bas and Öztürk-Işık used anatomical MRI from the UCSF-PDGM

dataset with deep learning and tumor-focused preprocessing [7]. Turhan et al. used ASL-based cerebral blood flow to classify IDH subgroups [9]. Our project follows this direction and focuses on contrast agent-free sequences, including ASL, together with a soft-mask pipeline.

3 Materials and Methods

3.1 Dataset and Labels

We use the University of California San Francisco Preoperative Diffuse Glioma MRI (UCSF-PDGM) dataset [2]. After selecting the cases used in this project, the dataset contains 495 patients. The IDH label is binary: IDH-wildtype (392 patients) or IDH-mutant (103 patients), so the classes are imbalanced. We split the patients into training, validation, and test sets using a single stratified 60/20/20 split with a fixed random seed, which keeps the same class ratio in each set. The test set contains 99 patients (21 IDH-mutant and 78 IDH-wildtype). Table 3.1 shows the split.

Table 3.1: Dataset split.

Set	Number of patients
Training	297
Validation	99
Test	99
Total	495

3.2 Preprocessing

The UCSF-PDGM images are already co-registered (the sequences are aligned to each other) and skull-stripped (non-brain tissue is removed). On top of this, we apply per-modality intensity normalization so that each sequence has a comparable range. We use all axial slices, resize each slice to 224×224 pixels, and stack the three sequences (T2w, FLAIR, ASL) as the three channels of the network input.

3.3 Model

Convolutional neural networks (CNNs) are deep learning models that work well on images. Training a CNN from scratch needs a lot of data, which is not available here, so we use transfer learning: we start from a model that was already trained on the large ImageNet dataset [5] and fine-tune it on our data.

We use a 2D ResNet-18 [3]. ResNet-18 is a convolutional network with 18 layers, organized into a first convolution and four stages of residual blocks. The residual (skip) connections make it easier to train a deeper network. After the residual stages, a global average pooling layer turns the feature maps into a 512-dimensional feature vector. The input

has three channels (T2w, FLAIR, and ASL) of size 224×224 . We replace the original 1000-class head with a small head made of a dropout layer and a single linear layer that outputs one value z (a logit) for the binary decision (IDH-wildtype vs. IDH-mutant). The logit is mapped to a probability with the sigmoid function,

$$p = \sigma(z) = \frac{1}{1 + e^{-z}}, \quad (3.1)$$

where $p \in [0, 1]$ is the predicted probability that the case is IDH-mutant. The model has about 11.2 million parameters and is trained end-to-end with the AdamW optimizer at a learning rate of 3×10^{-5} .

Figure 3.1 gives an overview of the approach, and the detailed network is shown in Figure 3.2. The full configuration is listed in Table 3.2, and we use the same configuration for all of the experiments in this report.

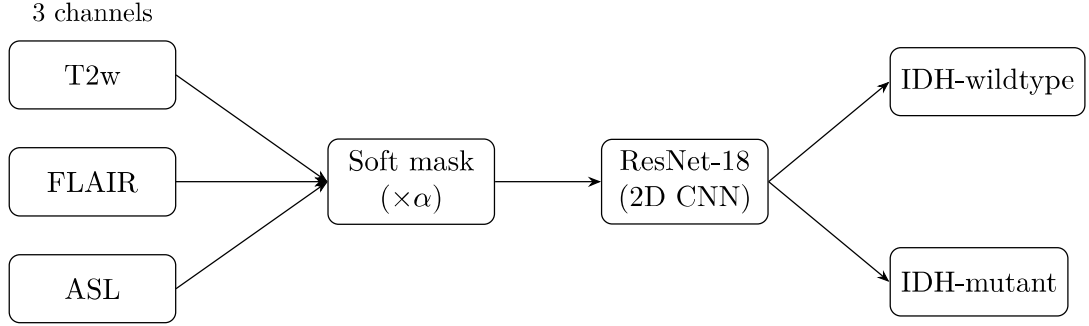


Figure 3.1: Overview of the pipeline: the three contrast agent-free sequences are soft-masked and passed to a ResNet-18 that predicts the IDH mutation status.

3.4 Soft-Mask Pipeline

We compare two pipelines that use the same data and the same model. In the no-mask pipeline, the three sequences are fed to the network as they are. In the soft-mask pipeline, each sequence is multiplied by a soft map $m = \alpha + (1 - \alpha)t$, where t is the tumor mask ($t = 1$ inside the tumor and 0 elsewhere). The tumor region therefore keeps its full intensity while the rest of the brain is scaled down by the factor α . This keeps the model focused on the tumor while still letting it see the surrounding brain, and it does not add an extra input channel (Figure 3.1). Figure 3.3 shows the effect of α on an example slice. We test several values of α to see how sensitive the result is to this choice.

3.5 Training and Patient-Level Decision

We train the network with a binary cross-entropy (BCE) loss. For a batch of N slices, with true label $y_i \in \{0, 1\}$ and predicted probability p_i , the loss is

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log p_i + (1 - y_i) \log(1 - p_i)]. \quad (3.2)$$

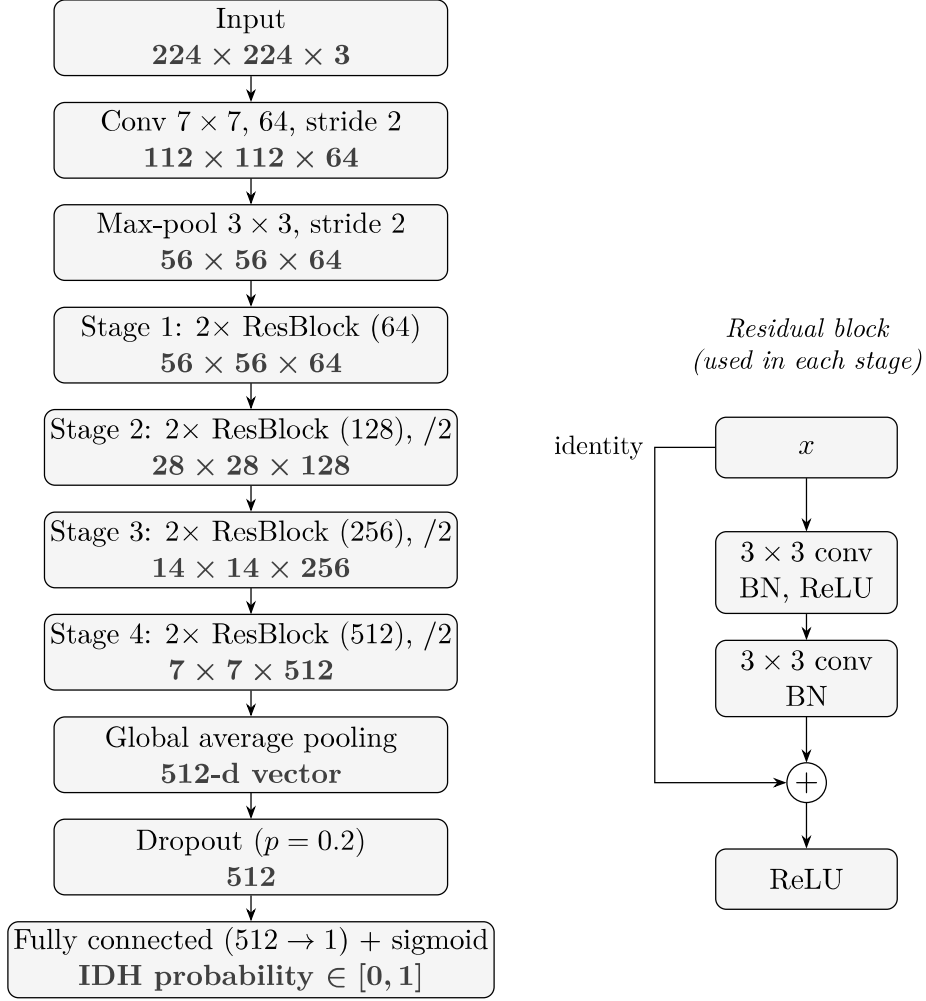


Figure 3.2: Detailed ResNet-18 architecture used in this work.

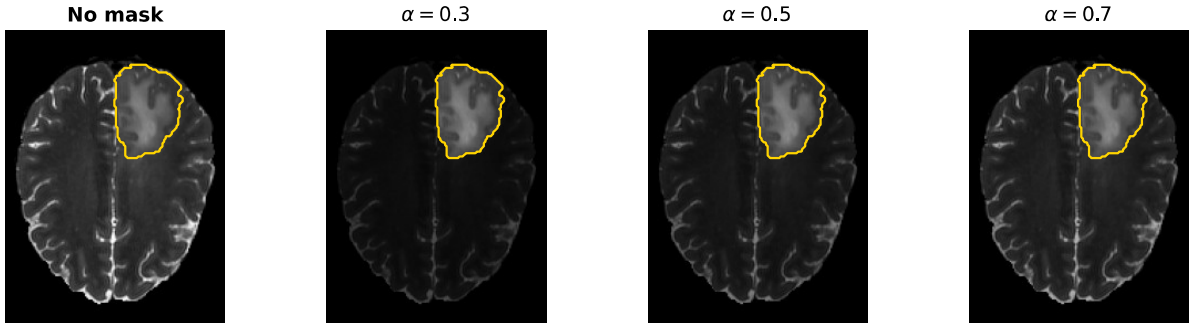


Figure 3.3: Effect of the soft mask on an example T2w slice. The tumor (outlined in yellow) keeps its full intensity while the rest of the brain is scaled by α . A smaller α dims the background more, and the no-mask pipeline corresponds to $\alpha = 1$.

We use the AdamW optimizer, and we apply data augmentation during training. The model makes a prediction for each slice. To get one prediction per patient, we take the mean of the slice probabilities over all of the patient’s slices. All choices are selected on the validation set, and the test set is used only once at the end.

Table 3.2: Model configuration and hyperparameters (used for all experiments).

Setting	Value
Backbone	ResNet-18 (ImageNet-pretrained)
Input	224×224 , 3 channels (T2w, FLAIR, ASL)
Classifier head	Dropout + linear layer (1 output)
Parameters	≈ 11.2 million
Optimizer	AdamW
Learning rate	3×10^{-5}
Weight decay	1×10^{-4}
Dropout	0.2
Epochs	40
Batch size	16
Loss	Binary cross-entropy
Augmentation	Flips, rotation ($\pm 20^\circ$), scaling, gamma, Gaussian noise
Patient-level decision	Mean of slice probabilities
Random seed	63

3.6 Experimental Design and Evaluation

We run three experiments:

1. Pipeline comparison: we compare the no-mask and soft-mask pipelines.
2. Soft-mask strength (α) sweep: we change α (0.3, 0.5, 0.7) to check how robust the result is.
3. ASL contribution: we retrain the soft-mask model on different subsets of the three sequences to measure how much each one, and ASL in particular, contributes.

We report standard classification metrics: the area under the ROC curve (AUC), F1 score, sensitivity (recall), specificity, precision, and accuracy. Taking IDH-mutant as the positive class, and writing TP, TN, FP, and FN for the numbers of true positives, true negatives, false positives, and false negatives at the chosen operating point, the threshold-based metrics are defined as

$$\begin{aligned}
 \text{Precision} &= \frac{\text{TP}}{\text{TP} + \text{FP}}, & \text{Recall (sensitivity)} &= \frac{\text{TP}}{\text{TP} + \text{FN}}, \\
 \text{Specificity} &= \frac{\text{TN}}{\text{TN} + \text{FP}}, & \text{Accuracy} &= \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}, \\
 \text{F1} &= \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.
 \end{aligned}$$

The area under the ROC curve (AUC) has no single-threshold formula: it is the area under the curve of the true-positive rate against the false-positive rate, and it equals the probability that the model gives a higher score to a randomly chosen IDH-mutant case than to a randomly chosen IDH-wildtype case. AUC is our main metric because it does not depend on a specific operating point. For each experiment, the best setting is chosen on the validation set, and the chosen model is then evaluated once on the test set.

4 Results

4.1 Primary Model

Table 4.1 shows the results of our best model on the test set: the soft-mask pipeline with $\alpha = 0.5$. It reaches an AUC of 0.987, an F1 score of 0.870, a sensitivity of 0.952, a specificity of 0.936, and an accuracy of 0.939. Out of the 21 IDH-mutant patients in the test set, the model correctly finds 20 (it misses 1), and it makes 5 false-positive predictions out of the 78 IDH-wildtype patients.

Table 4.1: Test-set results of the primary model (soft-mask, $\alpha = 0.5$).

Metric	Value
AUC	0.987
F1 score	0.870
Sensitivity (recall)	0.952
Specificity	0.936
Precision	0.800
Accuracy	0.939

4.2 Pipeline Comparison

Table 4.2 compares the no-mask and soft-mask pipelines, together with the three α values. The soft mask improves the AUC from 0.971 (no mask) to 0.987 ($\alpha = 0.5$). The improvement is consistent, but it is modest on this single split. Figure 4.1 shows the confusion matrix of the best configuration (soft-mask, $\alpha = 0.5$) on the test set.

Table 4.2: Pipeline and soft-mask strength (α) comparison on the test set.

Configuration	AUC	F1	Prec.	Recall	Spec.	Acc.
No-mask	0.971	0.851	0.769	0.952	0.923	0.929
Soft-mask, $\alpha = 0.3$	0.974	0.800	0.750	0.857	0.923	0.909
Soft-mask, $\alpha = 0.5$	0.987	0.870	0.800	0.952	0.936	0.939
Soft-mask, $\alpha = 0.7$	0.979	0.810	0.810	0.810	0.949	0.919

		Predicted	
		Mutant	Wildtype
Actual	Mutant	20 TP	1 FN
	Wildtype	5 FP	73 TN

Figure 4.1: Confusion matrix of the soft-mask ($\alpha = 0.5$) model on the test set.

4.3 Robustness to α

Table 4.2 also shows that the three α values (0.3, 0.5, 0.7) give similar AUCs (0.974, 0.987, and 0.979). This means the result is not very sensitive to the exact mask strength. We choose $\alpha = 0.5$ because it is the best on the validation set.

4.4 ASL Contribution

To measure the contribution of ASL, we retrain the soft-mask model on different subsets of the three sequences. Table 4.3 shows the results. When we remove ASL and use only T2w + FLAIR, the AUC drops from 0.987 to 0.939, so ASL adds about 0.048 AUC. ASL alone reaches an AUC of 0.966, which is higher than T2w + FLAIR together. This shows that ASL (perfusion) carries strong information about the IDH status, which fits the biology described in Section 2 and supports the contrast agent-free approach.

Table 4.3: ASL contribution (modality ablation) on the test set.

Sequences	AUC	F1	Prec.	Recall	Spec.	Acc.
T2w	0.939	0.757	0.875	0.667	0.974	0.909
FLAIR	0.918	0.698	0.682	0.714	0.910	0.869
ASL	0.966	0.809	0.731	0.905	0.910	0.909
T2w + FLAIR	0.939	0.722	0.867	0.619	0.974	0.899
T2w + ASL	0.971	0.818	0.783	0.857	0.936	0.919
FLAIR + ASL	0.967	0.780	0.800	0.762	0.949	0.909
T2w + FLAIR + ASL	0.987	0.870	0.800	0.952	0.936	0.939

4.5 Explainable AI (Grad-CAM)

To understand what the models look at, we use Grad-CAM, an explainable AI method that highlights the image regions that most affect the prediction [4]. Figure 4.2 compares the attention of the no-mask and soft-mask ($\alpha = 0.5$) models on several test cases. The maps show that both models focus on the tumor while still using the surrounding brain, and that the soft mask does not make the model rely only on the tumor region (it does not take a shortcut). This gives more confidence that the models use meaningful information.

Explainable AI is important in medical applications, so that clinicians can trust and check the model [8].

5 Conclusion

5.1 Results

This project showed that IDH mutation status can be predicted from contrast agent-free brain MRI. Our best model (soft-mask, $\alpha = 0.5$) reached a test AUC of 0.987, with a sensitivity of 0.952 and a specificity of 0.936. The soft mask gave a consistent improvement over the no-mask pipeline, and the ASL perfusion sequence was an important contributor.

5.2 Discussion and Future Work

These results are promising, but they have limitations. First, the dataset is relatively small and imbalanced, with many more IDH-wildtype than IDH-mutant cases. Second, the model architecture (ResNet-18) is not state-of-the-art. Third, the results come from a single train/validation/test split, so the exact numbers can change with a different split and should be seen as preliminary. Finally, the backbone is pre-trained on ImageNet, which is made of natural images. This can bring biases that are not ideal for medical images.

For future work, we plan to:

- use state-of-the-art architectures such as Vision Transformers (ViTs), ideally pre-trained on radiology data rather than natural images
- perform external validation on an independent dataset
- extend the analysis to volumetric (3D) data
- explore generative AI approaches

5.3 Realistic Constraints

5.3.1 Social, Environmental and Economic Impact

A contrast agent-free and non-invasive method has several benefits. For patients, it avoids the risks of a biopsy and the side effects of gadolinium contrast (which cannot be given to some patients), and it can give faster and more accessible results. For the environment, using less contrast agent reduces the amount of gadolinium that ends up in water systems. On the cost side, contrast agent-free MRI is cheaper than contrast MRI or surgery. At the same time, such a model should be used carefully: it is not a replacement for tissue diagnosis, and if the training data is not diverse, the model may be biased.

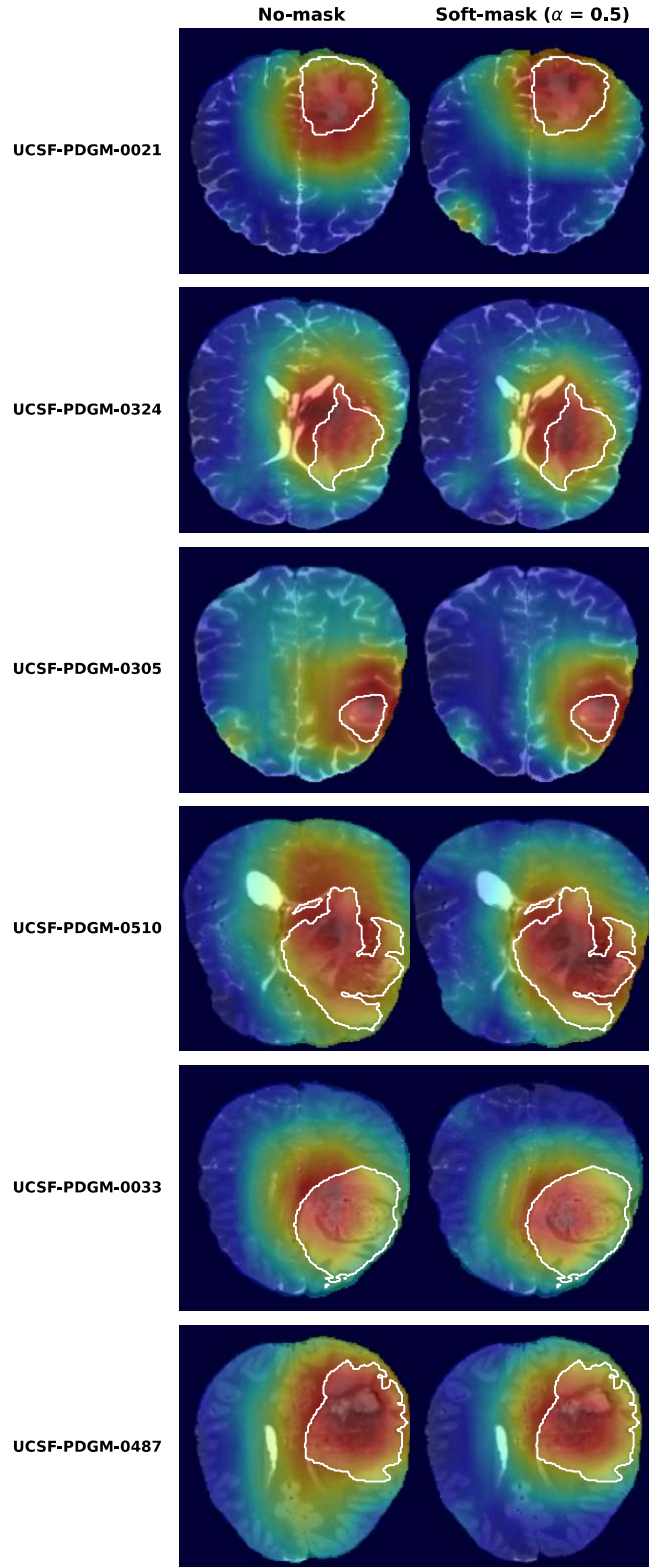


Figure 4.2: Grad-CAM attention for six representative test cases under the no-mask and soft-mask ($\alpha = 0.5$) models (tumor outlined in white). Both models concentrate on the tumor while still using the surrounding brain. The soft mask does not cause the model to rely on the tumor region alone.

5.3.2 Cost Analysis

The main cost of this project was the development time, which was about 250 hours over the term. The dataset (UCSF-PDGM) is publicly available at no cost, and all the software used is open source. Once trained, the model is cheap to run and does not need special hardware for prediction.

5.3.3 Standards

A real medical product based on this work would need to follow several standards and regulations. Medical images use the DICOM standard. The IDH labels follow the WHO CNS5 (2021) classification [1]. Medical device software is covered by standards such as IEC 62304 (software life cycle) and ISO 14971 (risk management). Patient data must be protected under privacy laws such as KVKK (Turkey), GDPR (EU), and HIPAA (US). To be used as a product, the model would also need regulatory approval, such as CE marking under the EU Medical Device Regulation (MDR) or clearance as Software as a Medical Device (SaMD) by the FDA.

APPENDICES

A Full Training Configuration

Table A.1 lists the complete configuration used for all experiments in this report, including the implementation details that were omitted from Table 3.2 in the main text for brevity.

Table A.1: Complete training configuration used for all experiments.

Setting	Value
<i>Architecture and input</i>	
Backbone	ResNet-18 (ImageNet-pretrained)
Input	224×224 , 3 channels (T2w, FLAIR, ASL)
Classifier head	Dropout (0.2) + linear layer ($512 \rightarrow 1$)
Parameters	≈ 11.2 million
<i>Optimization and schedule</i>	
Optimizer	AdamW
Base learning rate	3×10^{-5}
Layer-wise LR multipliers	$0.1 \times$ (conv1–layer3), $1.0 \times$ (layer4), $2.0 \times$ (head)
Weight decay	1×10^{-4}
LR schedule	10-epoch linear warm-up, then cosine decay
Epochs	40 (best by patient-level validation AUC)
Batch size	16
<i>Loss, balancing, and augmentation</i>	
Loss	Binary cross-entropy (label smoothing 0.1)
Class balancing	Weighted random sampler
Augmentation	Flips, rotation ($\pm 20^\circ$), scaling (0.85–1.15), gamma (0.8–1.2), Gaussian noise ($\sigma = 0.02$)
<i>Soft mask and inference</i>	
Soft mask	Tumor $\times 1$, background $\times \alpha$ (primary $\alpha = 0.5$)
Patient-level decision	Mean of per-slice probabilities
Operating point	F1-optimal (selected on validation)
<i>Data and reproducibility</i>	
Dataset	UCSF-PDGM v5 (495 patients)
Intensity normalization	Per-modality clip (0.5–99.5%), z-score, min–max [0, 1]
Slice selection	Axial slices with tumor area ≥ 250 voxels
Split	Stratified 60/20/20 (297 / 99 / 99), seed 63
Framework	PyTorch / torchvision

B Training Curves

Figure B.1 shows the training and validation loss and the patient-level validation AUC over the 40 epochs for the primary soft-mask ($\alpha = 0.5$) model. The validation curves

plateau without diverging from the training curves. The epoch with the highest validation AUC is the one used for evaluation.

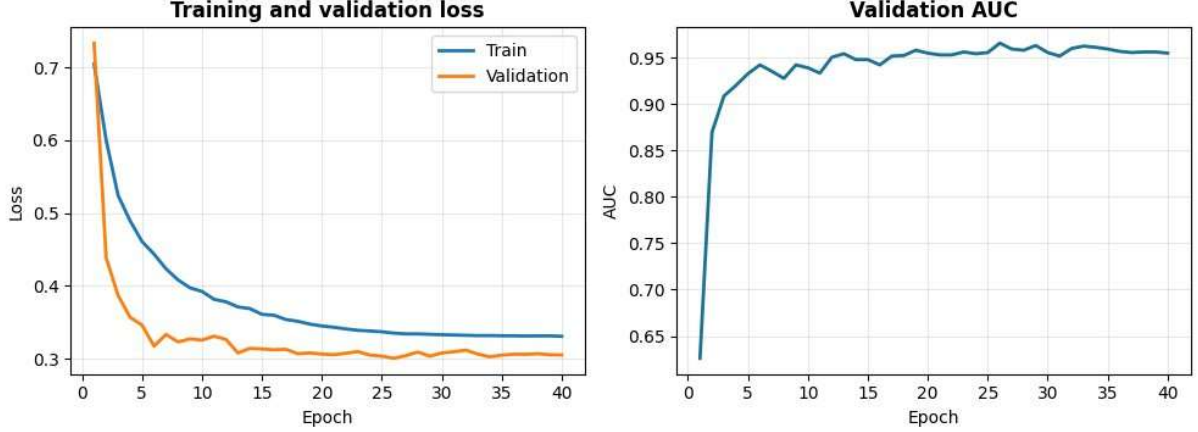


Figure B.1: Training and validation loss (left) and patient-level validation AUC (right) over training, for the soft-mask ($\alpha = 0.5$) model.

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Dedicated to all who fail and try, fail and try, fail and try...